WORKED SOLUTION

STEP III 1998, Question 8

Pure • Vector Geometry of Spheres

STEP III 1998, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q8

The key idea

Everything here reduces to the same move: write "the point $\mathbf{r}$ lies on the sphere" as a scalar equation $|\mathbf{r}-\mathbf{c}|^2 = a^2$, expand it with the dot product, and read off what you need. For the line in (i) this gives a quadratic in λ, and the number of intersections is decided by its discriminant. For (ii) we need the distance from the centre to the plane, and for (iii) two sphere equations subtracted from each other give exactly the dot product that "perpendicular" asks about.

Solution

Part (i)

The point $\mathbf{b}+\lambda\mathbf{m}$ lies on the sphere exactly when $(\mathbf{b}+\lambda\mathbf{m})\cdot(\mathbf{b}+\lambda\mathbf{m}) = a^2$. Expanding, and using $\mathbf{m}\cdot\mathbf{m} = 1$ because $\mathbf{m}$ is a unit vector,

$$\lambda^2 + 2(\mathbf{b}\cdot\mathbf{m})\lambda + \mathbf{b}\cdot\mathbf{b} - a^2 = 0.$$

Each real root λ gives one point of intersection, and different values of λ give different points (since $\mathbf{m} \neq \mathbf{0}$). So there are two points of intersection exactly when this quadratic has two distinct real roots, that is, when its discriminant is positive:

$$4(\mathbf{b}\cdot\mathbf{m})^2 - 4(\mathbf{b}\cdot\mathbf{b} - a^2) > 0 \iff a^2 > \mathbf{b}\cdot\mathbf{b} - (\mathbf{b}\cdot\mathbf{m})^2,$$

which is the required condition.

There is precisely one point of intersection when the discriminant is zero:

$$a^2 = \mathbf{b}\cdot\mathbf{b} - (\mathbf{b}\cdot\mathbf{m})^2.$$

Then the repeated root is $\lambda = -\mathbf{b}\cdot\mathbf{m}$ (minus half the coefficient of λ), so the point is

$$\mathbf{p} = \mathbf{b} - (\mathbf{b}\cdot\mathbf{m})\,\mathbf{m}, \qquad \text{and} \qquad \mathbf{m}\cdot\mathbf{p} = \mathbf{b}\cdot\mathbf{m} - (\mathbf{b}\cdot\mathbf{m})(\mathbf{m}\cdot\mathbf{m}) = 0,$$

again using $\mathbf{m}\cdot\mathbf{m} = 1$. So a tangent line is perpendicular to the radius to its point of contact, as it should be.

Part (ii)

A plane perpendicular to $\mathbf{n}$ has equation $\mathbf{r}\cdot\mathbf{n} = k$ for a constant k, and since $\mathbf{n}$ is a unit vector its distance from the origin is $|k|$. Choosing the sign of $\mathbf{n}$ so that $k \geq 0$ (that is, $\mathbf{n}$ points from the origin towards the plane), the plane is

$$\mathbf{r}\cdot\mathbf{n} = l.$$

Ask yourself How far is the centre $\mathbf{d}$ from this plane?

Go from $\mathbf{d}$ along the normal: the point $\mathbf{d}+\mu\mathbf{n}$ lies in the plane when $\mathbf{d}\cdot\mathbf{n}+\mu = l$, so $\mu = l - \mathbf{d}\cdot\mathbf{n}$. This is the foot of the perpendicular from $\mathbf{d}$, so the distance from the centre to the plane is $|\mu| = |l - \mathbf{d}\cdot\mathbf{n}|$.

The plane touches the sphere exactly when this distance equals the radius (a smaller distance means the plane cuts the sphere in a circle, a larger one means they miss). So the condition is

$$|\mathbf{d}\cdot\mathbf{n} - l| = a, \qquad \text{equivalently} \qquad (\mathbf{d}\cdot\mathbf{n} - l)^2 = a^2,$$

i.e. $\mathbf{d}\cdot\mathbf{n} = l \pm a$. (If instead $\mathbf{n}$ points away from the plane, its equation is $\mathbf{r}\cdot\mathbf{n} = -l$ and the condition reads $|\mathbf{d}\cdot\mathbf{n} + l| = a$.)

Part (iii)

The first sphere has centre O and the second has centre D (position vector $\mathbf{d}$), both of radius a. A point $\mathbf{r}$ lies on both when

$$\mathbf{r}\cdot\mathbf{r} = a^2 \qquad \text{and} \qquad (\mathbf{r}-\mathbf{d})\cdot(\mathbf{r}-\mathbf{d}) = a^2.$$

Expanding the second and subtracting the first gives $-2\,\mathbf{r}\cdot\mathbf{d} + \mathbf{d}\cdot\mathbf{d} = 0$, so every point of intersection satisfies

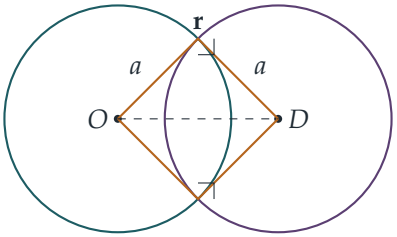
$$\mathbf{r}\cdot\mathbf{d} = \tfrac{1}{2}\,\mathbf{d}\cdot\mathbf{d}.$$

The two radii to the point $\mathbf{r}$ are the vectors $\mathbf{r}$ (from O) and $\mathbf{r}-\mathbf{d}$ (from D). They are perpendicular when

$$\mathbf{r}\cdot(\mathbf{r}-\mathbf{d}) = \mathbf{r}\cdot\mathbf{r} - \mathbf{r}\cdot\mathbf{d} = a^2 - \tfrac{1}{2}\,\mathbf{d}\cdot\mathbf{d}$$

is zero. The right-hand side does not depend on which intersection point we chose, so if $\mathbf{d}\cdot\mathbf{d} = 2a^2$ the radii are perpendicular at every point of intersection, which is what we had to show. (The argument also runs backwards: if the radii are perpendicular at even one common point, then $\mathbf{d}\cdot\mathbf{d} = 2a^2$.)

The spheres really do meet in this case: $|\mathbf{d}| = \sqrt{2}\,a$ lies strictly between 0 and $2a$, so they cross in a circle. The cross-section through O and D shows the right angles.



In the plane, two circles of radius a with perpendicular radii at a common point make OD the hypotenuse of an isosceles right-angled triangle, so $OD^2 = a^2 + a^2 = 2a^2$: the same answer by Pythagoras.

Checks

- In (i), take $\mathbf{b} = \mathbf{0}$: the line passes through the centre, the condition becomes $a^2 > 0$, and indeed such a line always meets the sphere twice.
- In (ii), with the sphere centred at the origin ($\mathbf{d} = \mathbf{0}$) the condition says $l = a$: a plane touches the sphere when it is at distance a from the centre.
- Numerically, with $a = 1$ and $|\mathbf{d}| = 1.2$ (not $\sqrt{2}$), an intersection point is $(0.6, 0.8, 0)$ and $\mathbf{r} \cdot (\mathbf{r} - \mathbf{d}) = 1 - 0.72 = 0.28 \neq 0$, matching $a^2 - \frac{1}{2}\mathbf{d} \cdot \mathbf{d}$. Random tests of all three parts agreed to within 10^{-13}.

Common slips

- Forgetting that $\mathbf{m} \cdot \mathbf{m} = 1$, and so carrying an unnecessary coefficient on λ^2.
- Writing the tangency condition in (ii) as $\mathbf{d} \cdot \mathbf{n} - l = a$ only: the centre can be on either side of the plane.

Beyond STEP

The quantity $\mathbf{b} \cdot \mathbf{b} - (\mathbf{b} \cdot \mathbf{m})^2$ is $|\mathbf{b} \times \mathbf{m}|^2$, the squared distance from the origin to the line, and $\mathbf{b} - (\mathbf{b} \cdot \mathbf{m})\mathbf{m}$ is the orthogonal projection of $\mathbf{b}$ away from $\mathbf{m}$, an idea at the heart of linear algebra. Spheres that cut at right angles are called orthogonal, and they are central to inversive geometry and to models of hyperbolic geometry. Strang, *Introduction to Linear Algebra*, covers projections; Needham, *Visual Complex Analysis*, shows orthogonal circles at work.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.